Subhajit Dandapat

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About me

Computational Astrophysicist working at the intersection of physics-informed machine learning, large-scale Bayesian inference, and AI-driven scientific discovery. My research focuses on weak-signal detection and uncertainty-aware modeling in high-dimensional, noise-dominated time-series data, with primary applications in gravitational-wave astronomy.

I have held collaborative and leadership roles in major international scientific consortia, including the International Pulsar Timing Array (IPTA), LIGO, and the ESA/NASA-funded LISA mission, contributed to the development of scalable inference frameworks, rigorous statistical validation methodologies, and interdisciplinary research coordination. I have demonstrated computational and funding leadership through securing major GPU/HPC allocations and contributing to competitive AI research grants.

My long-term vision is to advance physics-informed machine learning and foundation model paradigms that bridge academia and industry, translating first-principles scientific modeling into scalable, interpretable, and uncertainty-aware AI systems for complex scientific and applied research domains.

Education

2024–current **Postdoctoral Research Fellow**, *Department of Physics, National University of Singapore, Singapore*

Advisor: Dr. Alvin Chua

2019–2024 **Doctor of Philosophy in Astrophysics**, *Department of Astronomy & Astrophysics, Tata Institute of Fundamental Research, Mumbai, Maharashtra, India*

Thesis title: *Modeling Nano and Hecto Hertz Gravitational Wave sources and their observational implications*

Advisor: Prof. A. Gopakumar

2014–2019 **BS-MS Dual Degree in Physics, Indian Institute of Science Education and Research**, Bhopal, Madhya Pradesh, India

MS Thesis title: *Chiral Anomalies in Quantum Field Theory*

Studied: *Physics (Major), Mathematics, and Chemistry*

CGPA:8.89/10

Research & Professional Experience

Jan 2026-Present **IBM Research Collaborator :** Contributing to the development of a unified **foundation model** for scalable physics-informed machine learning, focused on scalable noise characterization, signal classification, and simulation-based Bayesian inference across diverse astrophysical GW systems.

Nov 2024-*present* **Postdoctoral Research Fellow**, *National University of Singapore*

- **Physics-Informed Transformer Architectures:** Designed and implemented a novel external-attention transformer with physics-informed positional encoding in PyTorch for detecting Eccentric supermassive black hole binaries (SMBHBs) in highly heterogeneous multimodal pulsar timing array (PTA) datasets. Reduces training epochs by a **factor of two** and **substantially reduces posterior uncertainty** by an **order of magnitude** relative to standard positional encodings. [In Prep.]
- **Amortized Simulation-Based Inference:** Leveraged the above physics-informed Transformer backbone to generate structured latent context representations for simulation-based Bayesian inference, conditioning discrete and continuous normalizing flows (DNFs, CNFs) for expressive amortized posterior density estimation in high-dimensional gravitational-wave parameter spaces. Reduced inference time from days to hours and significantly decreased reliance on computationally expensive Markov Chain Monte Carlo (MCMC) Bayesian pipelines through amortized neural inference, while maintaining accurate and uncertainty-calibrated results. [In Prep.]

- **Optimizing GW Signal Search with Physics-Informed Neural Networks (PINNs)** – Applied PINN surrogates to replace costly ODE/PDE solvers in International PTA pipelines for inspiraling, eccentric, spinning SMBHBs; accelerated Bayesian likelihood evaluations, reduced computational bottlenecks, and enabled scalable, hardware-accelerated GW data analysis.

2019-2024 **Research Scholar**, *Department of Astronomy & Astrophysics, Tata Institute of Fundamental Research, Mumbai, Maharashtra, India*

- **Bayesian Inference with Large-Scale MCMC** – Applied high-dimensional (several hundreds of parameters) MCMC Bayesian inference to extract weak GW signal parameters buried in heterogeneous, noise-dominated GW datasets, specifically pulsar timing array (PTA) datasets; enabled detection and characterization of supermassive black hole binaries (SMBHBs) from real observational data. It includes:
  - *PTAs:* GR post-Newtonian accurate waveform modeling of circular, eccentric, and hyperbolic BH binary orbits [gwecc_subhajit] [GW_hyp]; rigorous per-pulsar noise characterization and model selection; GWB characterization and spectral analysis across nanohertz frequency bands.
  - *LVK:* Developed `HyperbolicTD`, a LIGO Algorithm Library (LAL)-compatible 3PN accurate waveform model for GW signals from hyperbolic stellar-mass BH encounters [arXiv].
- **Cross-Collaboration Posterior Analysis** – Led quantitative comparisons of GW posteriors across multiple international PTA collaborations by evaluating distribution of their differences, moving beyond traditional point-based metrics (e.g., Mahalanobis distance) to more rigorous distribution-level comparisons. [Posterior_comparisons]
- **Burst-with-Memory Search Leadership** – Spearheaded the search for gravitational-wave burst-with-memory (BWM) signals in the International PTA collaboration, coordinating large-scale data analysis pipelines and cross-team validation efforts. [GWhyp]
- **Other Theoretical Efforts** – Extended 3PN accurate instantaneous contributions to radiated GW energy and angular momentum from hyperbolic passages of non-spinning compact objects, with implications for inspiral GWs in highly eccentric orbits [arXiv].

AI Grants Writing & Computing Resource Awards

- **Principal Author**, *CFP03 HPC Project Resource Allocation Award (NUS HPC, 9 Oct 2025 – Oct 2026)* — **Approved** computing allocation of **30K NVIDIA H100 GPU-hours**, **2K NVIDIA A40 GPU-hours**, and **200K CPU-hours** for the project *"AI- and GPU-Accelerated Search for Gravitational Waves in PTAs."*

- Co-authored **NRF Singapore AI-for-Science (AI4S) Catalytic Grant** – *"Unified AI Solutions to Enable Scientific Inference for Next-Generation Gravitational-Wave Astronomy"*. 05 September 2025, **1.5M SGD**, Singapore National Research Foundation.

- Contributed to **NAII Seed Grant** – *"Using Artificial Intelligence to Enhance Bayesian Inference in Gravitational-Wave Astronomy"*. 15 April 2025, **250k SGD**, NUS Artificial Intelligence Institute FY2025 H1 Seed Grant Call.

Selected Publications

Google Scholar: 2470+ citations | h-index: 14 | i10-index: 14 | 23 total publications

[1] Dandapat, S., Chua, A. J. (2026). *Transformers with physics-informed encodings and simulation-based inference for robust gravitational-wave detection in PTA data*. AI4X – Accelerate Conference 2026. [OpenReview]. [PDF].

[2] **Dandapat, S.** et al. 2024. *Efficient prescription to search for linear gravitational wave memory from hyperbolic black hole encounters and its application to the NANOGrav 12.5-year dataset.* Phys. Rev. D 109, 103018. arXiv:2402.03472.

[3] Agazie, G., **Dandapat, S.**, et al. 2024. *Comparing recent PTA results on the nanohertz stochastic GW background.* Astrophys. J. 966, 105. arXiv:2309.00693.

[4] **Dandapat, S.** et al. 2023. *Gravitational waves from black-hole encounters: Prospects for ground and galaxy-based observatories.* Phys. Rev. D 108, 024013. arXiv:2305.19318.

[5] Cho, G., **Dandapat, S.**, & Gopakumar, A. 2022. *Third order post-Newtonian gravitational radiation from two-body scattering: Instantaneous energy and angular momentum radiation.* Phys. Rev. D 105, 084018. arXiv:2111.00818.

[6] Antoniadis, J., ..., **Dandapat, S.**, et al. 2023. *The second data release from the European Pulsar Timing Array II. Customised pulsar noise models for spatially correlated gravitational waves.* A&A 678, A49. DOI:10.1051/0004-6361/202346842.

[7] Antoniadis, J., ..., **Dandapat, S.**, et al. 2023. *The second data release from the European Pulsar Timing Array III. Search for gravitational wave signals.* A&A 678, A50. DOI:10.1051/0004-6361/202346844.

Leadership Roles

2023 - present

- **PhD supervision**: Davendra (NUS); Liu Miaoxin (NUS); Vidit Singh (TIFR Mumbai); Adya Shukla (IIT Roorkee). **Graduate supervision**: Mark John (NUS). **Summer research supervision**: Arvind (Imperial College London). Projects involving gravitational-wave data analysis and AI-based approaches.
- Co-led the **IPTA Data Release 3 Search** for Gravitational Wave Burst with memory project.
- Co-led comparisons of posteriors across three major PTA collaborations (Arxiv:2309.00693; Arxiv:2306.16225 & Arxiv:2306.16214).

2022 - 2023

- Served as **Deputy Managing Leader (DML)** in Indian PTA, overseeing data reduction and backups for uGMRT datasets.

Skills

Programming Languages	`Python, Wolfram Language, Bash, Linux, C, LaTeX`
Machine Learning & AI	`PyTorch, Scikit-learn, NumPy, SciPy, Pandas, Normalizing Flows, PINNs, Transformer Simulation-based Inference`
Computing Software	`MATHEMATICA, MAPLE`
Astrophysical Software	`ENTERPRISE, BILBY, TEMPO2, PINT`
Telescope Observations	Upgraded Giant Metrewave Radio Telescope (uGMRT)
Data Analysis	Machine Learning, Bayesian Inference, Time-series Analysis, Data Visualization
HPC	Parallel Processing, GPU Computing (A40, H100), PBS/Slurm Environments

Fellowships and Awards

2025 NANOGrav **AccelNet** Travel Grant for international travel to attend conferences and research visits in the USA, June 15 - July 4, 2025.

2024 Received notable public visibility through a ***Times of India*** press release ([link]) highlighting the first-ever compelling evidence for nanohertz gravitational waves by the Indian initiative.

2023 **Sarojini Damodaran Fellowship** from TIFR supporting international travel to Europe for conferences and research visits.

2022 **Pauli Center for Theoretical Studies visitor program Fellowship**, ETH and University of Zürich, July 15 - September 17, 2022.

Seminars and Conference Presentations

2026

- **"Physics-informed features and simulation-based inference for gravitational-wave astronomy"**; **Invited talk**, Workshop for NUS–SNU Joint Symposium on *AI for Science*, **National University of Singapore**; February 11, 2026.

2025

- **"Searching for Linear GW Burst with memory in IPTA DR3 Dataset"**; In person talk on IPTA 2025 Science Meeting, **Caltech, Pasadena, US**; June 27, 2025.

2023

- **"Detection and Characterization of Burst with Linear Memory Events with Pulsar Timing Array and LIGO-Virgo-KAGRA Observatories"**; In person talk on Astrophysics group of Milan-Bicocca; December 5, 2023.
- **"Characterizing Burst with Linear Memory Events with LIGO-Virgo-KAGRA and Pulsar Timing Array Observatories"**; Amaldi15 virtual conference; July 17-21, 2023. [YouTube link]
- **"Searching Burst with memory event in IPTA dataset and Compairing PTA posteriors"**; IPTA Science meeting; 19-23 June, 2023.
- **"Comparing PTA posteriors"** IPTA GWA hackweek on 3P+ comparisons virtually on zoom; 6-8 March, 2023.

2022 **"Stochastic gravitational wave background spectrum due to supermassive black hole binaries in precessing eccentric orbits"**; Gravitational Wave Orchestra in-person held on UCLouvain, Belgium; Sep 8-9,2022.

2021

- **"Modeling GW burst with linear memory events"**; EPTA 2021 Winter meeting virtually on zoom Dec 6-8, 2021 and Astronomy Society of India meeting in-person held on **IIT Roorkee**; Mar 25-29, 2021.
- **"Effect of relativistic pericentre advance on the SGWB due to eccentric SMBH binaries"**; EPTA 2021 Summer meeting virtually on zoom; Apr 21-23, 2021.

Research Visits

2025 Visited Prof. Maura McLaughlin's group at West Virginia University from **30 th June to 3rd July, 2025** for IPTA collaborative work.

2023

- Visited Prof. Alberto Sesana's B Massive group at Milano-Bicocca from **4 th December to 8th December, 2023** for IPTA collaborative work.
- Visited Prof. Jetzer's group at University of Zürich from **20 th November to 1st December, 2023** for LVK collaborative work.

2022 Visited Prof. Jetzer's group at University of Zürich from **15 th July to 30th September, 2022** for LVK collaborative work.

References

- **Prof. Alvin Chua**
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- **Prof. A. Gopakumar**
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